

MTH204 (Section A)
Graded Worksheet–4

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

1. Solve the ODE

$$x^2 y'' - 3xy' + 10y = 0.$$

2. Solve the ODE

$$x^2 y'' + xy' + 9y = 0, \quad \text{given that } y(1) = 0, \ y'(1) = 2.5.$$

3. Find the solution of

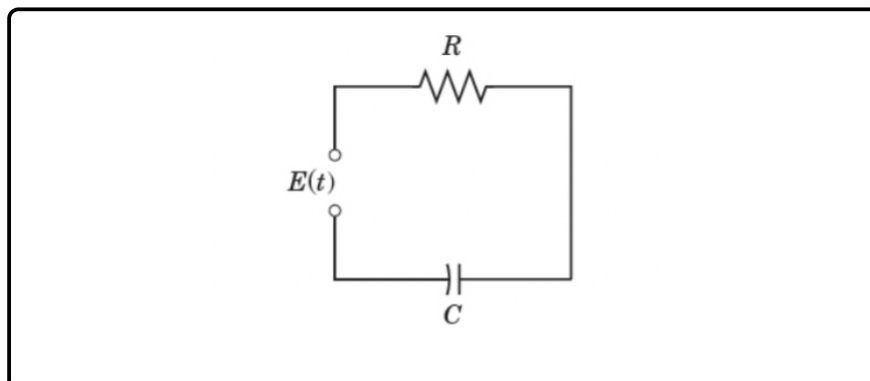
$$(a) \quad y'' + y' + \left(\pi^2 + \frac{1}{4}\right)y = e^{-x/2} \sin \pi x,$$

$$(b) \quad 8y'' - 6y' + y = 6 \cosh x, \quad \text{given that} \\ y(0) = 0.2, \ y'(0) = 0.05.$$

4. Find the transient motion of the mass-spring system modeled by the ODE

$$y'' + y = \cos \omega t, \quad \omega^2 \neq 1.$$

5. Model the RC circuit of the figure below.



- (a) Find the current due to a constant E .
 - (b) Find the current when $E = E_0 \sin \omega t$, and E_0 and ω are arbitrary.
6. Let $f(x)$ and $xf(x)$ be the particular solutions of a differential equation

$$y'' + R(x)y' + S(x)y = 0.$$

Then the solution of the differential equation

$$y'' + R(x)y' + S(x)y = f(x)$$

is

(a) $y = \left(-\frac{x^2}{2} + \alpha x + \beta\right) f(x)$

(b) $y = \left(\frac{x^2}{2} + \alpha x + \beta\right) f(x)$

(c) $y = (-x^2 + \alpha x + \beta) f(x)$

Solution:

Solution:

Solution:

Solution: